(number of eggs laid) was not controlled by the female's ability to lay eggs, since birds can lay many more eggs than they typically do. This fact can be demonstrated by replacement experiments, in which eggs are removed from a nest; the bird will re-lay repeatedly to compensate for the loss.

Instead, Lack said, the number of eggs laid by any species has evolved to fit with the food supply available. In other words, nature favors clutch sizes that correspond to the maximum number of young that the parents are likely to be able to sustain. So, if a pair of birds can only find enough food to feed six chicks, but the female has laid 12 eggs, those young will be hungry and may starve. If she has laid just one egg, although the chick will be raised successfully, most of the available food will have been unused. So neither the 12-egg nor the one-egg scenarios are good reproductive strategies; instead, laying six eggs offers the best chance of raising the most offspring.

This theory became known as the food limitation hypothesis, or Lack's principle, and it was later generalized by him and others to cover litter size in mammals and clutch size in fish and invertebrates.